Abhinav Yadav

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EXPERIENCE

Instahyre *New Delhi, India*

Senior Software Engineer (SDE III) 2024 - Present

- Owned technical direction across 5+ high-traffic backend domains (opportunity scoring, candidate ranking, recruiter workflows, notification pipelines, and background job orchestration) handling over 50M+ monthly API requests.
- Drove a system-wide latency reduction initiative via query refactoring, multi-layer caching, and targeted N+1 elimination; achieved 30-45% improvement in p95 response times across all high-traffic production API endpoints.
- Architected a zero-downtime Elasticsearch index versioning and live schema migration strategy across a 10M+ document corpus via alias hot-swap, automated rollback, and parallel reindex with zero impact on production traffic.
- Led AWS OpenSearch cluster migration: managed full index alias lifecycle, cross-version query compatibility, blue-green reindex pipelines, custom shard routing, CloudWatch health monitoring, and complete alerting configuration.
- Established org-wide engineering standards covering sync/async execution boundaries, cache invalidation contracts, and data ownership patterns; formally adopted as backend engineering guidelines across the entire engineering org.
- Served as designated P0/P1 on-call escalation owner; diagnosed and resolved systemic failures spanning Celery, RabbitMQ, Elasticsearch, and MySQL - reducing production incident frequency by approximately 40% over 18 months.
- Reviewed all performance-sensitive, security-critical, and data-integrity-impacting backend changes in the role of primary code review authority, enforcing correctness, reliability, and security standards across the backend org.
- Mentored 10+ engineers across junior to senior levels on production debugging, race condition analysis, migration design, async system patterns, and incident response; multiple mentees subsequently advanced to senior roles.

Senior Software Engineer (SDE II) 2022 - 2024

- Optimized high-traffic backend systems across Search, Candidates, Jobs, and Opportunities product areas, supporting 10K+ daily active users at peak concurrent load while maintaining consistent sub-200ms API response targets.
- Benchmarked Redis, ElastiCache, and Memcached under concurrent production-equivalent load scenarios; migrated from DatabaseCache to Redis with zero downtime, reducing steady-state database query load by approximately 35%.
- Designed Redis caching layer with per-key TTL management, safe cache invalidation strategies, and connection pooling; improved p95 API latency by 25-40% across the most frequently accessed and performance-critical endpoints.
- Designed and deployed a horizontally scalable multi-server architecture cleanly separating the web serving tier (uWSGI/NGINX), dedicated Celery worker pools, and outbound processing into independently deployable, scalable services.
- Resolved systemic Celery reliability failures: fixed retry storms, corrected queue misconfigurations, and refactored long-running tasks exceeding SLA time budgets, significantly and measurably increasing async throughput.
- Investigated and resolved production incidents involving RDS replica lag, InnoDB deadlocks, CPU saturation, and MySQL lock contention on high-write tables with millions of rows under sustained concurrent production load.
- Hardened production access controls with restricted SSH key policies, Unix user isolation, VPN-enforced internal routing, and full secrets lifecycle management through AWS Secrets Manager integration and automated key rotation.

Software Engineer (SDE I) 2019 - 2022

- Designed and shipped Employer Team Management featuring full RBAC (Admin/Member roles), a comprehensive audit trail, and impersonation safeguards; adopted by over 1,000 employer companies across the Instahyre platform.
- Identified and patched multiple authorization bypass vulnerabilities across backend API and frontend UI layers that exposed premium features to unauthorized users; hardened access control throughout the full backend request path.
- Resolved race conditions and non-atomic state transitions across Django ORM and Celery async workflows, eliminating an entire class of silent data corruption bugs that had persisted in production asynchronous job pipelines.
- Instrumented Datadog APM and Sentry across all API and async processing tiers; reduced mean time to detect (MTTD) production incidents by approximately 50% through structured alerting, tracing, and error rate aggregation.
- Tuned MySQL query execution plans via composite index design, carefully chosen covering indexes, and EXPLAIN-driven restructuring on multi-million-row tables, eliminating full-table scans on all performance-critical paths.

TECHNICAL SKILLS

Languages	Python, Go, Java, C/C++, JavaScript, SQL
Databases	MySQL (InnoDB), Redis, Elasticsearch / OpenSearch
Backend	Django, FastAPI, Spring Boot, Celery, RabbitMQ, Node.js, uWSGI, NGINX
Cloud	AWS (EC2, RDS, S3, CloudFront, Lambda, SNS, SQS, ElastiCache, IAM, CloudWatch)
Concepts	Distributed Systems, Microservices, Caching, Async Architecture, Performance Tuning
Observability	Datadog (APM, dashboards), CloudWatch, Sentry

EDUCATION

B.Tech, Computer Science - Dr. A.P.J. Abdul Kalam Technical University (AKTU) 2015 - 2019